

NANA KWAME OWUSU-AMPIAW

Electrical Engineering | Power Systems | Renewable Energy | Controls & Embedded Hardware
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EDUCATION

Bucknell University, College of Engineering - Lewisburg, PA Expected May 2028
Bachelor of Science in Electrical Engineering | GPA: 3.62/4.00
Relevant Coursework: Introduction to ECE Design, Circuit Theory, Foundations of Digital Systems, Electronics I, Differential Equations
Study Abroad: Management & Sustainability, University of Edinburgh, Scotland | Jun 2025

TECHNICAL SKILLS

Programming & Data: Python, MATLAB, Arduino, ThingSpeak
Electronics & Embedded: ESP32/Feather microcontrollers, sensor integration, circuit design, KiCad PCB layout, hardware debugging
Testing & Instrumentation: Analog Discovery 3, measurements & instrumentation, soldering, through-hole & surface-mount assembly
Energy & Controls: Microgrid concepts, solar energy analysis, energy monitoring, embedded control, automated systems
Fabrication: 3D printing, laser cutting, prototype assembly, hand & power tools

ENGINEERING EXPERIENCE

Thomas V. Giel Garage Doors LLC - Gibsonia, PA May 2026 - Aug 2026
Intern | Automated Systems & Field Operations

- Calibrated safety sensors, programmed operator controls, and configured travel limits for automated garage-door and screen systems to support reliable and safe operation.
- Assisted field technicians with installation, testing, and troubleshooting of automated garage doors, retractable awnings, and motorized screen systems.
- Supported field and warehouse operations using hand and power tools while coordinating equipment and materials for installations.

Bucknell Civil Engineering Department - Lewisburg, PA Jun 2025 - Aug 2025
Undergraduate Researcher

- Developed Python-based numerical models to simulate diffusion through porous media across varying time scales.
- Executed simulations and compared numerical results with analytical solutions to evaluate model behavior and accuracy.

ENGINEERING PROJECTS

Solar Tracking System - Bucknell University In Progress

- Developing a sensor-based control system that adjusts photovoltaic panel orientation in response to measured light intensity.
- Applying embedded control, sensor integration, and solar-energy concepts to photovoltaic panel positioning.

Smart Microgrid & Sustainable Home - Bucknell University Aug 2025 - Oct 2025

- Analyzed building energy data and contributed to a smart microgrid prototype incorporating automated appliance regulation, energy storage, solar-generation analysis, and grid sell-back concepts.
- Evaluated system performance and demonstrated a projected 15% reduction in energy consumption through automated energy-management strategies.

Embedded Sensing & IoT System - Bucknell University Aug 2025 - Dec 2025

- Developed an ESP32/Feather-based monitoring system integrating multiple sensors and visual indicators for real-time measurements.
- Integrated sensor wiring and data acquisition with ThingSpeak; performed hardware debugging, testing, and technical documentation.

ADDITIONAL TECHNICAL EXPERIENCE

Uptown & Weis Center, Bucknell University - Lewisburg, PA Feb 2025 - Present
Assistant Technical Manager, Uptown | Aug 2026 - Present
Technical Staff | Feb 2025 - Aug 2026

- Assist with managing student technical staff and technical operations for university events, including setup, troubleshooting, and maintenance of sound, lighting, and audiovisual equipment.
- Attend weekly managerial meetings and coordinate with performers, student staff, and technical managers to support reliable live-event operations and resolve technical issues.

LEADERSHIP & COMMUNITY ENGAGEMENT

Green Solutech - Accra, Ghana Sep 2023 - Oct 2023
Founder

- Founded and led a three-person team to build an Arduino-based contactless waste bin using an ultrasonic sensor and servo motor; taught Bishop Mixed JHS students how the system worked and lectured them on sustainability and e-waste awareness.

Peer2Peer Math Club, Prempeh College - Kumasi, Ghana Apr 2021 - Aug 2021
Co-Founder

- Co-founded a peer-tutoring initiative, mentored 60+ students in mathematics, and helped participating students improve test scores by approximately 25%.